

Creation Date 11-Nov-2011

Revision Date 16-Jul-2025

Revision Number 4

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Product Description: | <b>Lead foil, 2.0mm (0.08in) thick, Puratronic®</b> |
| Cat No. :            | <b>12047</b>                                        |
| Synonyms             | Lead metal                                          |
| Index No             | 082-014-00-7                                        |
| CAS No               | 7439-92-1                                           |
| EC No                | 231-100-4                                           |
| Molecular Formula    | Pb                                                  |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Reproductive Toxicity  
Effects on or via lactation  
Specific target organ toxicity - (repeated exposure)

Category 1A (H360FD)  
/ Effects on or via lactation (H362)  
Category 1 (H372)

### Environmental hazards

Chronic aquatic toxicity

Category 1 (H410)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H360FD - May damage fertility. May damage the unborn child  
H362 - May cause harm to breast-fed children  
H372 - Causes damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects

### **Precautionary Statements**

P201 - Obtain special instructions before use  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P308 + P313 - IF exposed or concerned: Get medical advice/attention

### **Additional EU labelling**

Restricted to professional users

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## **Section 3: Composition/information on ingredients**

### 3.1. Substances

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| Component | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                 |
|-----------|-----------|-------------------|----------|-----------------------------------------------------------------------------------|
| Lead      | 7439-92-1 | EEC No. 231-100-4 | <=100    | Repr. 1A (H360FD)<br>STOT RE 1 (H372)<br>Lact. (H362)<br>Aquatic Chronic 1 (H410) |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Lead      | -                                     | M = 10'  | -               |

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                                                                                                       |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Non-combustible. Do not allow run-off from fire-fighting to enter drains or water courses.

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## Hazardous Combustion Products

Lead, lead oxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1D

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

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## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component | European Union                   | The United Kingdom                                                      | France                                                         | Belgium | Spain                                          |
|-----------|----------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|---------|------------------------------------------------|
| Lead      | TWA: 0.15 mg/m <sup>3</sup> (8h) | STEL: 0.45 mg/m <sup>3</sup> 15 min<br>TWA: 0.15 mg/m <sup>3</sup> 8 hr | TWA / VME: 0.1 mg/m <sup>3</sup> (8 heures). restrictive limit |         | TWA / VLA-ED: 0.15 mg/m <sup>3</sup> (8 horas) |

| Component | Italy                                                       | Germany                                                                             | Portugal                            | The Netherlands                    | Finland                               |
|-----------|-------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Lead      | TWA: 0.15 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average | TWA: 0.004 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.032 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> 8 horas | TWA: 0.15 mg/m <sup>3</sup> 8 uren | TWA: 0.1 mg/m <sup>3</sup> 8 tunteina |

| Component | Austria                                                                                | Denmark                                                                        | Switzerland                                                                    | Poland                                  | Norway                                                                                                          |
|-----------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Lead      | MAK-KZGW: 0.4 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.1 mg/m <sup>3</sup> 8 Stunden | TWA: 0.05 mg/m <sup>3</sup> 8 timer<br>STEL: 0.1 mg/m <sup>3</sup> 15 minutter | STEL: 0.8 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.1 mg/m <sup>3</sup> 8 Stunden | TWA: 0.05 mg/m <sup>3</sup> 8 godzinach | TWA: 0.05 mg/m <sup>3</sup> 8 timer<br>STEL: 0.15 mg/m <sup>3</sup> 15 minutter. value calculated dust and fume |

| Component | Bulgaria                    | Croatia                                   | Ireland                                                                  | Cyprus                      | Czech Republic                                                                                                    |
|-----------|-----------------------------|-------------------------------------------|--------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Lead      | TWA: 0.05 mg/m <sup>3</sup> | TWA-GVI: 0.15 mg/m <sup>3</sup> 8 satima. | TWA: 0.15 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.45 mg/m <sup>3</sup> 15 min | TWA: 0.15 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 0.2 mg/m <sup>3</sup> biological test, toxic for reproduction |

| Component | Estonia                                                                                                      | Gibraltar                        | Greece                      | Hungary                                                                             | Iceland                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Lead      | TWA: 0.1 mg/m <sup>3</sup> 8 tundides. total dust<br>TWA: 0.05 mg/m <sup>3</sup> 8 tundides. respirable dust | TWA: 0.15 mg/m <sup>3</sup> 8 hr | TWA: 0.15 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> 8 órában. AK<br>TWA: 0.05 mg/m <sup>3</sup> 8 órában. AK | TWA: 0.05 mg/m <sup>3</sup> 8 klukkustundum. dust, fume, and powder<br>Ceiling: 0.1 mg/m <sup>3</sup> dust, fume, and powder |

| Component | Latvia                                                     | Lithuania                                                                                                   | Luxembourg                            | Malta | Romania                           |
|-----------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------|-------|-----------------------------------|
| Lead      | STEL: 0.1 mg/m <sup>3</sup><br>TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.15 mg/m <sup>3</sup> inhalable fraction IPRD<br>TWA: 0.07 mg/m <sup>3</sup> respirable fraction IPRD | TWA: 0.15 mg/m <sup>3</sup> 8 Stunden |       | TWA: 0.15 mg/m <sup>3</sup> 8 ore |

| Component | Russia                           | Slovak Republic                                                                                  | Slovenia                                                                                                          | Sweden                                                                                | Turkey                             |
|-----------|----------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------|
| Lead      | TWA: 0.05 mg/m <sup>3</sup> 1826 | TWA: 0.15 mg/m <sup>3</sup> inhalable fraction<br>TWA: 0.5 mg/m <sup>3</sup> respirable fraction | TWA: 0.1 mg/m <sup>3</sup> 8 urah inhalable fraction<br>STEL: 0.4 mg/m <sup>3</sup> 15 minutah inhalable fraction | TLV: 0.1 mg/m <sup>3</sup> 8 timmar. NGV<br>TLV: 0.05 mg/m <sup>3</sup> 8 timmar. NGV | TWA: 0.15 mg/m <sup>3</sup> 8 saat |

#### Biological limit values

List source(s):

| Component | European Union | United Kingdom | France | Spain | Germany |
|-----------|----------------|----------------|--------|-------|---------|
|-----------|----------------|----------------|--------|-------|---------|

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|      |  |  |                                                                                                                                                   |                                      |                                                 |
|------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------|
| Lead |  |  | Lead: 400 µg/L blood<br>Lead: 180 µg/L blood<br>indifferent sampling time<br>Lead: 300 µg/L blood<br>Lead: 200 µg/L blood<br>Lead: 100 µg/L blood | Lead: 70 µg/dL blood<br>not critical | Lead: 150 µg/L whole<br>blood (no restriction ) |
|------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------|

| Component | Italy                                    | Finland                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Denmark                     | Bulgaria                                                                                               | Romania                                                                                                                                                                                                                                                                                                                              |
|-----------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lead      | 60 Pb µg/100 mL blood<br>end of workweek | Lead: 1.4 µmol/L blood<br>time of day does not<br>matter.<br>Lead: 50 µg/dL blood . if<br>the medical examination<br>shows that the Lead<br>concentration in the<br>employee's blood is<br>higher than 50 µg/dL, he<br>must not be used for<br>work that involves<br>exposure to Lead<br>Lead: 40 µg/dL blood . if<br>the blood's Lead<br>concentration of even<br>one employee in the<br>workplace is 40 µg/dL or<br>more, the employer<br>must especially monitor<br>the Lead concentration<br>in the air of the<br>workplace, the Lead<br>concentration in the<br>employees' blood and<br>the possible health<br>hazards caused by Lead | Lead: 20 µg/100 mL<br>blood | Lead: 300 µg/L blood<br>not fixed for women<br>under 45 years old<br>Lead: 400 µg/L blood<br>not fixed | Lead: 150 µg/L urine<br>end of shift<br>Lead: 70 µg/100 mL<br>blood end of shift<br>Lead: 3 mg/cm hair end<br>of shift<br>.delta.-Aminolevulinic<br>acid: 10 mg/L urine end<br>of shift<br>Coproporphyrin: 300<br>µg/L urine end of shift<br>free Erythrocytes<br>protoporphyrin: 100<br>µg/100 mL Erythrocyte<br>blood end of shift |

| Component | Gibraltar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Latvia                                                                                                                         | Slovak Republic                                                                                                                                                                                                                                                                                                                                  | Luxembourg                                                                                                                                                                                                                                                                      | Turkey                      |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Lead      | 70 µg/100 mL blood<br>Lead binding biological<br>limit value;biological<br>monitoring must include<br>measuring the<br>blood-lead level using<br>absorption spectrometry<br>or a method giving<br>equivalent results<br>0.075 mg/m³ air 40<br>hours per week Lead<br>medical surveillance<br>must be carried<br>out;threshold measured<br>in individual employees<br>40 µg/100 mL blood<br>Lead medical<br>surveillance must be<br>carried out;threshold<br>measured in individual<br>employees | Lead: 30 µg/100 mL<br>blood<br>Coproporphyrin: 100<br>µg/g Creatinine urine<br>Aminolevulinic acid: 5<br>mg/g Creatinine urine | Lead: 400 µg/L blood<br>not critical<br>Lead: 100 µg/L blood<br>not critical women<br>younger than 45 years<br>of age<br>.delta.-Aminolevulinic<br>acid: 15 mg/L urine not<br>critical<br>.delta.-Aminolevulinic<br>acid: 6 mg/L urine not<br>critical women younger<br>than 45 years of age<br>Coproporphyrins: 0.30<br>mg/L urine not critical | Lead: 70 µg/100 mL<br>blood.<br>Lead: 0.072 mg/m³<br>blood. medical<br>surveillance threshold in<br>air measured as a time<br>weighted average over<br>40 hours per week<br>Lead: 40 µg/100 mL<br>blood. medical<br>surveillance threshold<br>measured in individual<br>workers | Lead: 70 µg/100 mL<br>blood |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

MDHS6/3 Lead and inorganic compounds of lead in air Laboratory method using flame or electrothermal atomic absorption spectrometry

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

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No information available

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                   | Fresh water    | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)         |
|-----------------------------|----------------|--------------------------------|--------------------|------------------------------------|----------------------------|
| Lead<br>7439-92-1 ( <=100 ) | PNEC = 2.4µg/L | PNEC = 186mg/kg<br>sediment dw |                    | PNEC = 100µg/L                     | PNEC = 212mg/kg<br>soil dw |

| Component                   | Marine water   | Marine water sediment          | Marine water Intermittent | Food chain               | Air |
|-----------------------------|----------------|--------------------------------|---------------------------|--------------------------|-----|
| Lead<br>7439-92-1 ( <=100 ) | PNEC = 3.3µg/L | PNEC = 168mg/kg<br>sediment dw |                           | PNEC = 10.9mg/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

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## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Solid                    |                                   |
| Appearance                              | Grey                     |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | 327.4 °C / 621.3 °F      |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 1740 °C / 3164 °F        | @ 760 mmHg                        |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | Insoluble                |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | 1.77 mmHg @ 1000 °C      |                                   |
| Density / Specific Gravity              |                          |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

### 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Pb                     |
| Molecular Weight  | 207.19                 |
| Evaporation Rate  | Not applicable - Solid |

## Section 10: Stability and reactivity

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Exposure to air. Incompatible products.

### 10.5. Incompatible materials

Strong acids. Ammonium nitrate: fertilizers capable of self-sustaining decomposition. Peroxides.



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## 10.6. Hazardous decomposition products

Lead, lead oxides.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

|            |                   |
|------------|-------------------|
| Oral       | No data available |
| Dermal     | No data available |
| Inhalation | No data available |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

|             |                          |
|-------------|--------------------------|
| Respiratory | No data available        |
| Skin        | No data available        |
|             | No information available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC     |
|-----------|----|----|---------|----------|
| Lead      |    |    |         | Group 2A |

##### (g) reproductive toxicity; Reproductive Effects

Category 1A  
May cause harm to the unborn child. Possible risk of impaired fertility.

(h) STOT-single exposure; No data available

##### (i) STOT-repeated exposure;

|               |                                              |
|---------------|----------------------------------------------|
| Target Organs | Kidney, Central nervous system (CNS), Blood. |
|---------------|----------------------------------------------|

##### (j) aspiration hazard;

Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available.

### 11.2. Information on other hazards

#### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

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## Section 12: Ecological information

### 12.1. Toxicity

#### Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                                                                                                          | Water Flea                         | Freshwater Algae |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------|
| Lead      | LC50: = 1.32 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 1.17 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 0.44 mg/L, 96h semi-static (Cyprinus carpio) | EC50: = 600 µg/L, 48h (water flea) |                  |

| Component | Microtox | M-Factor |
|-----------|----------|----------|
| Lead      |          | M = 10'  |

### 12.2. Persistence and degradability

#### Persistence Degradability Degradation in sewage treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Insoluble in water, May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

### 12.4. Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

### 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

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|                                       |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b>         | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>              | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                                                                                                       |
| <b>Switzerland - Waste Ordinance</b>  | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## Section 14: Transport information

### IMDG/IMO

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| <b>14.1. UN number</b>                  | UN3077                                             |
| <b>14.2. UN proper shipping name</b>    | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. |
| <b>Technical Shipping Name</b>          | Lead                                               |
| <b>14.3. Transport hazard class(es)</b> | 9                                                  |
| <b>14.4. Packing group</b>              | III                                                |

### ADR

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| <b>14.1. UN number</b>                  | UN3077                                             |
| <b>14.2. UN proper shipping name</b>    | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. |
| <b>Technical Shipping Name</b>          | Lead                                               |
| <b>14.3. Transport hazard class(es)</b> | 9                                                  |
| <b>14.4. Packing group</b>              | III                                                |

### IATA

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| <b>14.1. UN number</b>                  | UN3077                                             |
| <b>14.2. UN proper shipping name</b>    | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. |
| <b>Technical Shipping Name</b>          | Lead                                               |
| <b>14.3. Transport hazard class(es)</b> | 9                                                  |
| <b>14.4. Packing group</b>              | III                                                |

|                                    |                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.5. Environmental hazards</b> | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|------------------------------------|----------------------------------------------------------------------------------------------------------|

|                                           |                                  |
|-------------------------------------------|----------------------------------|
| <b>14.6. Special precautions for user</b> | No special precautions required. |
|-------------------------------------------|----------------------------------|

|                                                                      |                                |
|----------------------------------------------------------------------|--------------------------------|
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods |
|----------------------------------------------------------------------|--------------------------------|

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **International Inventories**

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No | EINECS | ELINCS | NLP | IECSC | TCSI | KECL | ENCS | ISHL |
|-----------|--------|--------|--------|-----|-------|------|------|------|------|
|-----------|--------|--------|--------|-----|-------|------|------|------|------|

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|           |           |           |                                               |   |     |      |          |       |       |
|-----------|-----------|-----------|-----------------------------------------------|---|-----|------|----------|-------|-------|
| Lead      | 7439-92-1 | 231-100-4 | -                                             | - | X   | X    | KE-21887 | X     | -     |
|           |           |           |                                               |   |     |      |          |       |       |
| Component | CAS No    | TSCA      | TSCA Inventory notification - Active-Inactive |   | DSL | NDSL | AICS     | NZIoC | PICCS |
| Lead      | 7439-92-1 | X         | ACTIVE                                        |   | X   | -    | X        | X     | X     |

Legend: X - Listed '-' - Not listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Lead      | 7439-92-1 | -                                                                   | Use restricted. See entry 72.<br>(see link for restriction details)<br>Use restricted. See entry 30.<br>(see link for restriction details)<br>Use restricted. See entry 63.<br>(see link for restriction details)<br>Use restricted. See entry 75.<br>(see link for restriction details) | SVHC Candidate list - 231-100-4 - Toxic for reproduction (Article 57c)                                |

### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

## Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Lead      | 7439-92-1 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

| Component                  | ANNEX I - PART 1<br>List of chemicals subject to export notification procedure (referred to in Article 8) | ANNEX I - PART 2<br>List of chemicals qualifying for PIC notification (referred to in Article 11) | ANNEX I - PART 3<br>List of chemicals subject to the PIC procedure (referred to in Articles 13 and 14) |
|----------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Lead<br>7439-92-1 ( ≤100 ) | sr — severe restriction<br><br>i(2) — industrial chemical for public                                      | -                                                                                                 | -                                                                                                      |

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

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Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                |
|-----------|---------------------------------------|--------------------------------------------------------|
| Lead      | nwg                                   | Class II : 0.5 mg/m <sup>3</sup> (Massenkonzentration) |

| Component | France - INRS (Tables of occupational diseases)     |
|-----------|-----------------------------------------------------|
| Lead      | Tableaux des maladies professionnelles (TMP) - RG 1 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                   | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Lead<br>7439-92-1 ( <=100 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H360Df - May damage the unborn child. Suspected of damaging fertility

H362 - May cause harm to breast-fed children

H372 - Causes damage to organs through prolonged or repeated exposure

H410 - Very toxic to aquatic life with long lasting effects

H360FD - May damage fertility. May damage the unborn child

H360Fd - May damage fertility. Suspected of damaging the unborn child

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

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**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

11-Nov-2011

## Revision Date

16-Jul-2025

## Revision Summary

SDS sections updated.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**